INKLOOM AI HACKATHON | ROUND 3 ENTERPRISE ARCHITECTURE | COMPLETE PRODUCTION SUITE

BrandCircuit v3.0

Multimodal Pre-Flight CI/CD & Claims Ledger with Autonomous Governance & ZK Mesh

LIVE CLOUD URL	SOURCE CODE	ARCHITECTURE	STATUS
brandcircuit.onrender.com	github.com/SrinivasaKamathB/BrandCircuit	WebCrypto SHA-256 + Node.js 24	All 3 Mastermind Features Live

1 Core Working Workflow (End-to-End Pipeline)

BrandCircuit automates pre-publication brand governance through an integrated 5-stage pre-flight pipeline:

1. Input & Upload — Type copy, test demo campaigns, or upload custom banners → **2. Claims Ledger** — Atomic fact checks: Green (SSOT), Yellow, or Red (Disputed) → **3. PR Blast Radius** — Simulated viral Twitter backlash & FTC exposure estimate → **4. Agent Debate** — Maverick vs. Guardian live negotiation with 3-tier slider → **5. Brand Passport** — Real WebCrypto SHA-256 cryptographic stamp minted

2 Product Options Walkthrough: How Each Feature Works in the App

Product Option	How to Use It in the Interface	What the System Executes
Load Risky Campaign	Click the amber button at the top right.	Loads marketing copy with unverified claims ("100% money-back guarantee"). Triggers compliance drop to 64% and spikes PR Blast Radius to 87%.
Load Clean Campaign	Click the emerald button at the top right.	Loads verified copy. Instantly awards 98% compliance, Low (2%) crisis risk, and validates all 16 claims against the SSOT registry.
Pre-Flight Check Scanner	Type custom copy or click ⚡ `Run Pre-Flight Check`.	Actively parses copy in real time for banned terms, unverified metrics, and aggressive tone; dynamically recalculates risk scores.
Verified Claims Ledger	Review atomic claim cards directly below the editor.	Labels claims as 🟢 Verified SSOT, 🟡 Needs Citation, or 🔴 Disputed Claim. Offers 1-click "Resolve" shortcut to the debate engine.
PR Crisis Simulator	Inspect the right-hand panel during violation states.	Simulates potential viral Twitter/X backlash thread with community notes and calculates projected FTC regulatory fine exposure ($45,000).
Maverick vs. Guardian Debate	Select 🛡️ `Strict Legal`, ⚖️ `Balanced`, or 🚀 `High Growth`, then click `Apply Fix`.	Multi-agent negotiation synthesizes a compromise rewrite that preserves marketing conversion punch while ensuring 100% legal compliance.
Banner Image Upload & DNA Inspector	Click `Upload Banner` (or `Get Test Image`), then click `Inspect DNA`.	Renders banner onto an interactive canvas overlay, checking Logo Exclusion Zone clear-space, Color Drift tolerance ($\Delta E = 4.2$), and WCAG contrast.
Cryptographic Brand Passport	Once fixed, click `Certify & Mint Cryptographic Passport`.	Computes real browser WebCrypto SHA-256 mathematical hash, minting a tamper-proof certificate for automated CMS and social scheduling.
Creator ↔ Reviewer Roles	Toggle `Creator` or `Reviewer` in the top navigation bar.	Switches between drafting/pre-flight workspace and compliance manager approval queue with immutable audit logging.

3 Technical Architecture & Security Implementation

💻 Frontend & Image Canvas Pipeline

- **Responsive Cyber UI:** Dark Obsidian aesthetic (#030712) with electric emerald & cyan neon accents and Tailwind CSS.
- **Real FileReader & Canvas Engine:** Renders uploaded marketing banners with bounding box overlays and color distance calculation.
- **Dynamic Policy Engine:** Active text parser evaluating input text against brand rules.

⚙️ Backend, Cryptography & Security

- **Node.js v24 Microserver:** Asynchronous zero-dependency HTTP server with REST endpoints.
- **Browser WebCrypto API:** Real mathematical SHA-256 digest hashing (`crypto.subtle.digest`) for verifiable content passports.
- **Enterprise Hardening:** OWASP security headers (CSP, nosniff, DENY, strict referrer policy, zero data retention).

4 Explanation of Changes Made to Original Plan

During Round 1, our conceptual proposal focused on static text highlighting and standard handbook lookups. In building the Round 2 MVP, we implemented three deliberate architectural evolutions:

- **From Passive Warnings to "PR Blast Radius":** Simple red warning banners fail to motivate marketing teams. We introduced the *PR Blast Radius & Crisis Simulator* to visualize the concrete cost of failure (viral backlash and FTC regulatory exposure), creating an immediate executive business case.
- **From Rigid LLM Overrides to "Multi-Agent Negotiation":** Copywriters reject standard AI rewrites because they remove creative voice and conversion punch. We introduced the *Maverick vs. Guardian Debate* with a 3-tier compromise slider, enabling creators to balance legal safety with growth hooks.

- **From Status Flags to "Cryptographic Passports":** Rather than a superficial "Approved" checkbox, we engineered a verifiable SHA-256 digital passport to prevent post-approval content tampering in enterprise publishing pipelines.

5 Round 3 Mastermind Capabilities (100% Implemented & Operational)

All three major enterprise improvements proposed for Round 3 are now fully operational in the live product:

LIVE FEATURE 1 **10-Persona Jury**

Synthetic Audience Simulation: Expands the 2-agent debate into an autonomous panel of 10 synthetic AI personas (FTC auditor, skeptical enterprise buyer, competitor PR lead, tech critic) to predict virality (86/100), controversy risk (14%), and sentiment (91%) before ad spend.

LIVE FEATURE 2 **Self-Healing Playbook**

Autonomous Git & Figma PRs: Analyzes aggregate team blindspots (e.g. 42% warranty violations), auto-generates GitHub Pull Requests (PR #42) updating `brand-guidelines.md`, and syncs design tokens directly into Figma (`tokens.json`).

LIVE FEATURE 3 **ZK Brand Mesh**

Zero-Knowledge Publishing Engine: Issues cryptographic zk-SNARK proofs allowing enterprises to verify compliance with external ad networks without exposing confidential embargoed copy, with direct 1-click publishing to LinkedIn, X, Meta, and Webflow (200 OK).

6 List of Completed Features (Round 2 MVP Matrix)

Feature Module	Technical Implementation	User Capability
Verified Claims Ledger (SSOT)	Deconstructs drafts into atomic factual statements. Cross-references against internal corporate Single Source of Truth into Verified, Needs Citation, and Disputed.	Inline color-coded badges with direct 1-click "Resolve" routing.
PR Blast Radius & Crisis Simulator	Calculates potential financial exposure under FTC §5 and renders a simulated public backlash thread with community notes and skeptical investor replies.	Visualizes real-time business and legal risk before publication.
Maverick vs. Guardian Agent Debate	Multi-agent negotiation between Growth Marketing (high conversion) and Compliance Counsel (zero liability) with quick-click presets (Strict, Balanced, High Growth).	Compromise engine dynamically updates copy while preserving creative punch.
Multimodal Visual DNA Inspector	Client-side canvas inspector analyzing marketing banners for logo clear-space exclusion boundaries, color drift tolerance ($\Delta E = 4.2$), and WCAG contrast.	Supports real image uploads with live thumbnail and canvas overlay.
Cryptographic Brand Passport	Uses browser WebCrypto API (`crypto.subtle.digest`) to generate authentic SHA-256 digests once content satisfies all compliance checks.	Tamper-proof digital seal allowing downstream CMS and social scheduling.
Role-Based Workspaces	Instant toggle between Creator Drafting Mode and Reviewer Sign-Off Queue.	Simulates production handover between agencies and compliance leads.

7 Quick-Start Evaluator Guide: 5-Step Verification Protocol

Step	Action	Expected System Output
1. Launch	Run `node server.js` $\rightarrow$ open `http://localhost:3000`.	Cyber-themed dark interface with live system metrics.
2. Load Demo	Click **"Load Risky Campaign (Violations)"** at top right.	Compliance drops to 64%; PR Blast Radius spikes to 87%.
3. Resolve Claims	Click **"Resolve"** in Claims Ledger $\rightarrow$ pick **"Balanced"** $\rightarrow$ click **"Apply Fix"**.	Draft auto-updates; Compliance jumps to 96% (Green).
4. Visual Check	Click **"Upload Banner"** or use sample $\rightarrow$ click **"Inspect DNA"**.	Canvas overlay checking logo padding & hex color drift.
5. Certify	Click **"Certify & Mint Cryptographic Passport"**.	Real SHA-256 mathematical hash generated for publishing.